

1. Create a table comparing at least three types of operating systems (such as Windows, Linux, and Android) based on their main features, target devices, and examples of apps you use on each.

Operating System	Main Features	Target Devices	Examples of Apps
Windows	User-friendly interface, multitasking, wide hardware/software support, strong gaming support	Desktops, laptops, tablets	Microsoft Office, Chrome, Zoom, VLC
Linux	Open-source, highly customizable, secure, stable, supports command-line tools	Desktops, laptops, servers, embedded systems	Firefox, LibreOffice, VLC, GIMP
Android	Touchscreen interface, app ecosystem, notifications, Google services, customizable	Smartphones, tablets, Smart TVs	WhatsApp, Instagram, YouTube, Google Maps

2. Identify the operating system running on your phone and your laptop/desktop. Write down two unique features of each OS that affect your daily use of popular apps like Instagram or Zomato.

Device	Operating System	Two features	Effects on Daily App use
Phone	Android	1. Notifications- Android allows apps to send real-time notifications.	Instagram can notify me about messages and likes, while Zomato can

			send order and delivery updates.
Phone	Android	2. App permissions- I can control access to camera, location, microphone and storage.	Instagram can use the camera for photos/videos, while Zomato can use location to find nearby restaurants and track delivery.
Laptop/Desktop	Windows 10	1. Multitasking - Windows supports multiple apps and windows at the same time.	I can use Instagram in a browser while working on documents or watching videos.
Laptop/Desktop	Windows 10	2. File Management- File explorer makes it easy to organize and access downloaded files.	I can download images, receipts or invoices from websites such as Zomato and easily find them later.

- 3. List three security risks that are specific to mobile operating systems like Android or IOS, and explain how these risks could impact apps you use (for example, WhatsApp or Paytm).**

Three Mobile Operating System Security Risks

Security Risks	Explanation	Impact on Apps
1. Malicious Apps & Malware	Fake or unsafe apps may contain malware that steals data or monitors activity.	Malware could steal WhatsApp messages, contacts, or Paytm login details and financial information.

2. Excessive App Permissions	App may request unnecessary access to the camera, microphone, contacts, SMS or location.	A malicious app with SMS access could potentially read OTPs used for Paytm transactions.
3. Unsecured Wi-Fi and Network Attacks	Connecting to public or fake Wi-Fi networks can expose mobile traffic to attackers.	Attackers may try to interrupt login information or other sensitive data when using apps such as WhatsApp or Paytm.

4. Research and describe how a cloud-based operating system (such as Chrome OS or a virtual desktop environment) is different from a traditional desktop OS. Give one example of how these difference changes the way you access apps like YouTube or Flipkart.

Cloud-Based Operating System vs Traditional Desktop OS

Feature	Clout-Based OS	Traditional Desktop OS
Where apps run	Mainly web/cloud-based apps	Mainly locally installed applications
Storage	Storage focus on cloud storage such as Google Drive	Strong focus on local SSD/HDD with optional cloud storage.
Internet dependence	Many features work best with an internet connection although some apps support offline use	Most installed applications can run without internet.
Updates	ChromeOS and its web environment are designed to stay automatically up to date.	Updates are managed through the OS and individual applications
Hardware requirement	Generally lightweight and designed for simple, web-	Usually requires more powerful hardware for

	focused computing	desktop software
Examples	ChromeOS on a Chromebook	Windows 10 on a laptop/PC

Example

- Suppose you are using a Chromebook with ChromeOS. TO use YouTube or Flipkart, you can simple open chrome and visit the website. The application is primarily accessed through the web rather than installing a traditional Windows program.
- For example, In ChromeOS – Open Chrome/YouTube.com/Watch videos
- And in Windows, Open browser/app/ YouTube/Watch videos